3. HEALTHSAT - Decentralized Disease Outbreak Prediction

This mission focuses on a multispectral satellite constellation designed to monitor environmental factors such as water quality and flooding that are linked to disease outbreaks. By integrating AI and blockchain, the system replaces traditional, reactive surveillance with an early warning network where governments and health organizations can share predictive models and coordinate responses through decentralized protocols to break down existing data silos.

Satellite Requirements

- **Orbit:** Sun-synchronous LEO 680 km, 4-satellite constellation for daily global coverage
- **Coverage:** Priority - tropical regions (malaria, dengue, cholera zones), urban flood areas
- **Payload:**
    - Multispectral imager (visible + NIR + SWIR + thermal: 8 bands, 10m resolution)
    - Water vapor sensor (atmospheric moisture - predicts mosquito breeding)
    - Thermal IR (temperature anomalies, urban heat islands)
- **Power:** 65W average (moderate imaging power)
- **Mass:** <120 kg per satellite
- **Comms:** S-band 20 Mbps downlink, daily ground station passes

Web3 System

Data Flow: Satellite captures environmental data (standing water after floods, vegetation indices, temperature) → Onboard AI correlates with disease outbreak patterns → Uploads risk maps to IPFS → Smart contract records risk scores per geographic region → Health clinics/local health workers report actual disease cases → AI models retrain on confirmed data → High-risk alerts trigger: vaccine distribution, mosquito control, water treatment → All actions + outcomes recorded on-chain for transparent accountability → Communities earn tokens for accurate disease reporting

Key Smart Contracts:

1. **EnvironmentalData** - Stores satellite-derived risk indices (water presence, temperature, NDVI)
2. **DiseaseRegistry** - Decentralized disease case reporting by health workers (privacy-preserving)
3. **PredictiveModel** - On-chain AI model parameters updated with new training data
4. **EarlyWarningSystem** - Automatically alerts regions when risk crosses threshold
5. **ResponseCoordination** - Health orgs claim responsibility for interventions, report outcomes
6. **IncentivePool** - Rewards health workers for accurate case reporting, punishes false data

Round 1 constraints:

1. Mission Objectives & Success Criteria

Every design must start with the **"Why."** In a hackathon, your mission objective isn't just a goal; it's a boundary.

- **Operational Lifetime:** How long must the satellite survive? (e.g., 6 months for a tech demo vs. 5 years for a weather satellite).
- **Target Orbit:** This dictates almost everything else.
    - **LEO (Low Earth Orbit):** Lower latency, higher resolution, but high atmospheric drag.
    - **GEO (Geostationary Orbit):** Constant coverage of one area, but extremely high launch costs and radiation.
- **Data Latency:** Does the data need to be "real-time" (for disaster relief) or can it be stored and forwarded once a day?

2. Payload Constraints

The payload is the "customer" of the satellite; everything else (power, thermal, comms) exists just to keep it happy.

- **SWaP-C:** This is the golden rule of satellite design: **S**ize, **W**eight, **a**nd **P**ower - **C**ost.
- **Mass & Volume:** Most hackathons use the "U" standard (1U = 10x10x10 cm). You'll likely be restricted to a 3U or 6U form factor.
- **Duty Cycle:** Can the payload run 24/7, or does it need to shut down to recharge batteries while the satellite is in the Earth's shadow?

3. Budget & Resource Estimation

In a real-world scenario (and a good hackathon), "unlimited money" isn't a thing.

- **Total Cost of Ownership (TCO):** You must account for:
    1. **Development:** Components and labor.
    2. **Launch:** Often calculated per kilogram (e.g., $50,000/kg for LEO).
    3. **Operations:** Ground station fees to talk to your satellite.
- **COTS vs. Custom:** Using **Commercial Off-The-Shelf (COTS)** parts is cheaper and faster but might not be "space-hardened" against radiation compared to custom, radiation-hardened components.

4.Technical Constraints and Requirements

1. Environmental Surveillance & Sensing

- **Multispectral Detection:** The constellation utilizes **Thermal Infrared (TIR)** and **Short-Wave Infrared (SWIR)** sensors. Unlike standard cameras, these detect specific moisture levels and heat signatures to identify stagnant water (mosquito breeding grounds), algae blooms, and flood-prone zones.
- **Edge-AI Processing:** To solve the problem of "data-to-decision" delay, the satellites process imagery **on-board**. Instead of downloading massive image files, the AI identifies anomalies

(e.g., a 20% increase in stagnant water) and sends a lightweight "High Priority" alert to Earth immediately.

2. The Predictive Data Engine

- **Zoonotic Spillover Modeling:** The system doesn't just watch for disease; it predicts it. By tracking environmental shifts—like deforestation or flooding—the AI models how animal migrations might bring new viruses closer to human populations.
- **The Risk Atlas:** This is a real-time, global map that layers environmental data with population density. It gives health organizations (like the WHO or NGOs) a "weather forecast" for potential outbreaks before the first person gets sick.

3. Blockchain & Decentralized Protocol

- **Data Integrity Ledger:** In many regions, health data is censored or hidden for political reasons. By recording satellite telemetry on a **blockchain**, the environmental record becomes "immutable" (unchangeable). No government can "scrub" the data to hide a looming crisis.
- **Automated Response Triggers:** We use **Smart Contracts** as emergency switches. If the satellite detects environmental risk levels exceeding a pre-set threshold, the contract can automatically unlock emergency funds or trigger the dispatch of medical supplies to that specific GPS location.
- **Decentralized Data Marketplace:** This allows local doctors and researchers to contribute "ground-truth" data (like actual clinic test results) to improve the satellite's accuracy. In return, they earn tokens or governance rights in the network.

4. Governance & Integration

- **Silo-Breaking:** Traditional health data is trapped in old, disconnected government databases. HEALTHSAT provides an **Integration Toolkit**—software "adapters" that allow these old systems to securely plug into our decentralized network.
- **Early Warning SOPs:** The mission includes pre-negotiated **Standard Operating Procedures**. This means that when the satellite says "Risk Level: High," the response plan (vaccine deployment, water treatment) is already agreed upon and starts immediately.